

ControlledRAG: A Seven-Axis Reporting Standard for RAG Faithfulness Claims

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Abstract

RAG faithfulness claims are often reported from one retriever, one generator, one scorer, one threshold, and one dataset. We argue that such single-cell numbers are *under-identified*: any collapsed axis can materially change the reported headline under another defensible audit choice. We propose **ControlledRAG**, a seven-axis reporting standard covering generator, retriever, context structure, scorer, human calibration, threshold transfer, and cost, and demonstrate it through a self-audit of a previously published refinement-paradox claim about retrieval coherence. On the same fixed generations, NLI backbones sign-flip when moved from a zero-shot label proxy to a context-conditioned calling convention, with disjoint bootstrap CIs on a 600-row balanced subset. Scorer rankings against adjudicated human labels also change across slices: RAGAS-style leads on a typical-row calibration ($n=99$), while **roberta-large-mnli** leads on a targeted disagreement slice ($n=100$) and the legacy DeBERTa proxy turns weakly negative. A matched-context CCS intervention is null (-0.002 , paired Wilcoxon $p=0.628$). The anonymized review artifact includes per-query CSVs, both human-evaluation slices, source traces, and analysis scripts. ControlledRAG turns RAG faithfulness from a single-score claim into an auditable, seven-axis standard.

1 Introduction

Retrieval-augmented generation (RAG) [Lewis et al., 2020] now powers deployed systems across medical QA, legal research, and customer support, yet many papers still evaluate faithfulness with a narrow template: one retriever, one generator, one scorer, one threshold, and one benchmark. That template is convenient, but it *under-identifies* the claim. A result that looks decisive in one generator/retriever/scorer cell can change sign, shrink, or become a deployment tradeoff under an equally defensible audit choice.

The trigger for this paper is a previously published refinement-paradox claim of the form “improving per-passage retrieval reduces answer faithfulness.”¹ The headline was interesting enough to audit, but it does not survive the controls reported here. At scale, the original SQuAD generator-7B contrast shrinks to 0.661/0.650/0.661 across no-refinement, aggressive-refinement, and gated-refinement conditions; a matched-similarity intervention on context coherence is null; threshold-transfer cells change sign; and a cost-aware head-to-head produces no dominating system.

The audit reframes the original headline as a stronger methodological result. The problem is not that a coherence hypothesis “failed”; the problem is that single-cell RAG faithfulness numbers hide too many degrees of freedom. The empirical results in this paper are not independent anecdotes: each one corresponds to an axis that a RAG faithfulness claim must disclose.

The positive contribution is **ControlledRAG**: a reusable audit checklist and prescriptive minimum reporting standard for RAG faithfulness claims. ControlledRAG is not a new retriever, not a new faithfulness metric, and not a method leaderboard. It asks authors to vary or explicitly hold fixed seven axes—generator, retriever, context structure, scorer, human calibration, threshold transfer, and cost—so reviewers can tell whether a claim is robust or merely a metric, retriever, generator, threshold, or cost artifact.

Contributions. We make four contributions, mapped to the seven-axis standard. (i) **The standard itself (§4)**: a seven-axis disclosure checklist for RAG faithfulness claims, with one principle—each axis is a movable

¹Cited at submission as a public claim; the preprint reference is omitted here for double-blind review and will be added on acceptance.

quantity that, if left unreported, lets a reader substitute a different reasonable value and reach a different conclusion. (ii) **A self-applied seven-axis audit** (§5–§10) on a previously published refinement-paradox claim, exercising every axis of the standard. (iii) **Two completed adjudicated human-eval slices** (§8): a typical-row calibration ($n=99$, $\kappa=0.774$) and a targeted disagreement slice ($n=100$, $\kappa=0.721$) that reach *different* scorer rankings; the slice-dependence is the load-bearing finding under the human-calibration axis. (iv) **An anonymized review artifact** containing auditable per-query CSVs, matched context pairs, both human-eval slices, source traces, and analysis scripts.

Why NeurIPS. ControlledRAG is methodological, not NLP-specific. The under-identification phenomenon is a measurement-theory issue for any LLM-judged evaluation: same fixed generations, different reasonable scorers, different reported contrasts. The seven-axis decomposition is the contribution. We *conjecture* the decomposition generalises to other LLM-as-judge settings (summarisation faithfulness, code-execution evaluation, chain-of-thought scoring); demonstrating that generalisation is out of scope here.

Roadmap. §2 compresses related work. §3 describes the pipeline and audited cells. §4 defines ControlledRAG. §5–§10 run the seven-axis audit. §11 discusses what survives. §12 states limitations. The supplement contains pre-specification (§1), per-seed and per-cell detail (§2–§9), the context-conditioned NLI rescoring (§7), the completed $n = 99$ calibration (§12) and the completed $n = 100$ targeted disagreement evaluation (§13), the source trace (§14), and the related-work positioning table (§16).

2 Background and Related Work

A standard RAG system [Lewis et al., 2020, Gao et al., 2024] threads a dense retriever [Karpukhin et al., 2020, Wang et al., 2024], an optional reranker [Nogueira and Cho, 2019], an optional refinement step (e.g., Xu et al. [2024]), and a generator. The implicit assumption is that better local retrieval yields more faithful answers globally. Decoder-side correctors take a different position: Self-RAG [Asai et al., 2024] retrains the generator to emit reflection tokens, and CRAG [Yan et al., 2024] adds a retrieval evaluator that triggers web search; both pay latency cost. RAPTOR [Sarthi et al., 2024] clusters retrieved passages into a tree and retrieves from both leaves and LLM-summarised internal nodes (§10). Faithfulness is typically scored with NLI-style support metrics or LLM judges [Honovich et al., 2022, Laban et al., 2022, Es et al., 2024], occasionally with human annotation. Recent evaluation frameworks (ARES [Saad-Falcon et al., 2024], RAGChecker [Ru et al., 2024], RAGTruth [Niu et al., 2024], MIRAGE [Park et al., 2025], FaithJudge [Tamber et al., 2025]) supply judges, corpora, or benchmark suites; ControlledRAG is the disclosure layer around them. Discourse coherence has a long history in classical generation [Li and Hovy, 2014] but has not been quantified as a deployment-time RAG signal (further coherence references in supplement §16). Shi et al. [2023] and Liu et al. [2024] document distraction by irrelevant context and lost-in-the-middle behaviour respectively; we treat the Context Coherence Score (CCS) and answer-bearing evidence presence as two complementary context-structure variables to *report* alongside per-passage similarity, not as substitutes for human evaluation or cross-scorer checks. The supplement reproduces a positioning table mapping each prior framework to the audit axis it most directly engages.

3 Setup and Audited Cells

We hold the entire RAG pipeline fixed across audit experiments and vary only the audit axis under test. The audited cells are deliberately narrow—primarily short-answer extractive QA on SQuAD and PubMedQA with 7B-class generators—so that scorer, retriever, and generator choices can be cleanly separated.

Datasets. The matched-context, scaled, metric-fragility, and span-control experiments use SQuAD [Rajpurkar et al., 2016]; the threshold-transfer and noise experiments additionally use PubMedQA [Jin et al., 2019], HotpotQA [Yang et al., 2018], NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA. The long-form stress test on QASPER and MS-MARCO is supplement-only and not treated as broad generalization evidence (§12 and supplement).

Pipeline. Embeddings come from `sentence-transformers/all-MiniLM-L6-v2`; indexing uses ChromaDB; refinement uses `cross-encoder/ms-marco-MiniLM-L-6-v2`. Default chunk size is 1024 characters with 10% overlap and default $top-k$ is 3. Generators are served via local Ollama (Mistral-7B, Qwen2.5-7B). The matched-context audit also re-encodes the same 200 pairs with BGE-large and E5-large.

Conditions. Three retrieval configurations act as instruments for measuring the audit surface, not as systems to crown a winner. We name them by function and (in parentheses) by the prior naming convention: **no-refinement** ($top-k$ dense retrieval, also called *baseline*); **aggressive-refinement** (refine every retrieved passage through sub-chunking and cross-encoder selection; the prior-work name is *HCPC-v1* and this is the original failure-mode condition the published refinement paradox was based on); and **gated-refinement** (a coherence-gated variant with top -rank protection, a refinement cap, and merge-back; prior-work name *HCPC-v2*). Tables and section text use the functional names on first reference and the short prior names thereafter.

Metrics. Per-query **faithfulness** in the legacy audit tables is the stored automated support score in $[0, 1]$; **hallucination** indicates score < 0.5 unless a table states another label. The frozen DeBERTa and `roberta-large-mnli` scores are sentence-level zero-shot entailment-label proxies (answer sentence classified against entailment / neutral / contradiction with `hypothesis_template="{}`"; the retrieved context is retained in the per-row data but not consumed by either proxy). The RAGAS-style judge is context-conditioned by construction: a local Mistral-7B receives the question, retrieved context, and answer and returns a support score on $[0, 1]$. **Retrieval similarity** is the mean cosine similarity between the query and each retained chunk. **CCS** is the mean minus the standard deviation of off-diagonal pairwise cosine similarities between retained chunk embeddings (same embedder). The answer-span indicator is a lowercased, punctuation-stripped lexical match of any normalised gold-answer alias in the retrieved context.

Statistical protocol. The matched-context test is pre-specified with paired Wilcoxon and a 10,000-resample bootstrap CI on the mean paired difference; the scaled audit reports seed-level paired Wilcoxon over five seeds; binary rates use Wilson 95% CIs. P-values outside the pre-specified matched test are descriptive across multiple audit cells; we do not treat exploratory $p < 0.05$ rows as standalone discoveries. All audit cells, paired contrasts, and per-query CSVs are in the anonymized review artifact.

4 ControlledRAG: The Minimum Reporting Standard

ControlledRAG is a checklist. A paper that makes a RAG faithfulness claim fills in seven entries, one per axis, and a reviewer reads those entries to decide whether the claim is generator-universal, scorer-dependent, threshold-dependent, or otherwise narrow. The reporting-checklist template ships with the artifact as `controlledrag_reporting_checklist.md` (a filled-in copy for this paper is included; this is *distinct* from the mandatory NeurIPS paper checklist that ships at the end of the submitted PDF). Each entry asks: what was held fixed, what was varied, why this choice, and what is the known sensitivity to alternative reasonable values? The checklist is not a leaderboard rubric and does not score papers; it is a disclosure layer that a reviewer can use to ask one targeted question per axis.

Why seven axes, not six or eight. A movable degree of freedom belongs on the checklist if leaving it unreported lets a reader substitute a different reasonable value and reach a different conclusion. The seven axes the checklist asks every RAG faithfulness paper to report, with the overclaim each blocks, are: (1) *generator* (model family, size, decoding, fixed vs. regenerated outputs) blocks generator-universal claims; (2) *retriever* (embedder, index, reranker, refinement policy) blocks “retrieval quality” as one variable; (3) *context structure* (CCS, answer span, mean similarity) blocks CCS-as-oracle; (4) *scorer* (scorer names, input format, thresholds, cross-scorer correlations) blocks single-score claims (§7); (5) *human calibration* (label protocol, raters, agreement, and scorer-alignment correlations on at least one typical-row slice *and* on any targeted disagreement-driven slice) blocks treating any metric ranking as settled (§8); (6) *threshold transfer* (tuned τ , tune dataset, off-diagonal recovery table) blocks reusable-threshold claims (§9); (7) *cost* (per-condition latency, indexing time, external-call assumptions, including baselines) blocks leaderboards

without cost (§10). Six axes drop one of these movable quantities; eight start to double-count (decoding is a generator subfield; index size is a retriever subfield) without adding a new degree of freedom. The remainder of the paper supplies a self-applied audit in which each empirical cell motivates one axis.

5 The Matched-Context Audit

The matched-context audit asks the diagnostic question directly: *at fixed mean per-passage query–passage similarity, does higher CCS predict higher faithfulness?* The null answer is not the end of the analysis. It creates a constructive diagnostic question: what variables distinguish answer support once mean similarity is fixed?

Construction. For SQuAD with Mistral, we form 200 HIGH/LOW pairs so that mean query–passage similarity is matched to within 0.02 (0.4236/0.4215) while CCS is large in HIGH and small in LOW (0.670/0.137).

Result. Table 1 reports the pre-specified HIGH–LOW contrast. The mean paired difference is -0.002 , paired Wilcoxon $p = 0.628$, Cohen’s $d_z = -0.017$, 10,000-resample bootstrap 95% CI $[-0.022, +0.017]$. The HIGH hallucination rate is also *higher* than the LOW rate (16.5% vs. 9.0%). The pre-specified H_1 (HIGH>LOW faithfulness) is not supported in the audited cell. The continuous mean contrast and the thresholded hallucination contrast measure different parts of the score distribution, so a null mean can coexist with threshold-sensitive tail differences. The matched test rules out the simple HIGH-CCS-improves-faithfulness story in this audited cell.

Table 1: Matched-context audit on SQuAD/Mistral ($n=200$ pairs, matched mean query–passage similarity).

Quantity	HIGH-CCS	LOW-CCS	Difference	Test
Mean similarity	0.4236	0.4215	+0.0021	matched by construction
Mean CCS	0.670	0.137	+0.533	—
Faithfulness	0.636	0.639	-0.002	paired Wilcoxon $p = 0.628$
Hallucination	16.5%	9.0%	+7.5 pp	McNemar $p = 0.011$
95% CI on faith diff	—	—	$[-0.022, +0.017]$	10,000-resample bootstrap

Span presence in matched contexts. A lexical gold-answer-string check finds the answer more often in LOW-CCS contexts than HIGH-CCS contexts (27.0% vs. 17.0%, McNemar $p = 0.011$). The structural property captured by CCS is therefore not the same as “the retrieved context contains the answer”; the lexical span check is a weak proxy because retrieved text can contain the answer string while failing to support the generated answer.

Answer-span/control diagnostic. A regression and classification panel on the matched cell ($n = 400$ contexts; full table in supplement §3) shows that for continuous DeBERTa faithfulness, answer-span presence is the strongest single predictor (R^2 ranking $0.019 > 0.003 > 0.002$), while for binary hallucination AUROC, CCS is the strongest single predictor ($0.639 > 0.586 > 0.531$). The joint feature set has the highest score on both ($R^2 = 0.025$, AUROC = 0.643). Span and CCS therefore diagnose complementary failure modes; we do not claim a directional role for either, that span subsumes CCS, or that CCS subsumes span.

6 The Scaled Audit

The scaled audit asks the same headline question—does aggressive-refinement (HCPC-v1) reliably hurt faithfulness on SQuAD/Mistral?—at $n = 2,500$ examples per condition with five seeds, each analysed by paired Wilcoxon on 500-pair contrasts. Pooled means are 0.661/0.650/0.661 for no-refinement / aggressive-refinement / gated-refinement with overlapping Wilson CIs on hallucination rate (14.6%/14.7%/14.3%; Table 2). One of five seeds reaches $p < 0.05$ for the no-refinement–aggressive-refinement contrast (seed 44,

paired $p = 0.008$); the other four yield $p \in [0.083, 0.234]$. The gated–aggressive contrast reaches significance in 3 of 5 seeds with small per-seed magnitude (0.003–0.019).

Table 2: Scaled audit on SQuAD/Mistral, $n = 2,500$ per condition, pooled across five seeds. Conditions are functional names with prior naming in parentheses.

Condition	Faithfulness (95% CI)	Hallucination (Wilson)	Retrieval sim.	Refine rate
no-refinement (baseline)	0.661 [0.655, 0.667]	14.6% [13.3, 16.0]	0.532	0.000
aggressive-refinement (HCPC-v1)	0.650 [0.645, 0.656]	14.7% [13.4, 16.2]	0.569	1.000
gated-refinement (HCPC-v2)	0.661 [0.655, 0.667]	14.3% [13.0, 15.7]	0.532	0.705

The originally published $n = 30$ headline ($0.7995 \rightarrow 0.6407$ on no-refinement vs. aggressive-refinement; see footnote 1) does not survive at scale. The scaled contrast is small and seed-sensitive and should not be cited as a broad refinement-hurts-faithfulness claim. This is itself a ControlledRAG-style audit lesson: at $n = 30$ the contrast looked clean enough to report as a phenomenon, but at $n = 2,500$ it does not.

7 Fixed-Generation Metric Fragility

This section is the core empirical pillar for the scorer axis. For the scaled audit’s 7,500 Mistral rows, outputs are fixed before rescoreing, retrieval condition is fixed, and the only changed component is the scorer. The three scorers are not on a common numerical scale—two are legacy zero-shot label proxies that do not consume the retrieved context, and the third is a context-conditioned local judge—so raw score differences across scorers are not interpreted as directly comparable causal estimates. To address the scale concern, we standardize each scorer over the same fixed rows by both z-scoring and rank-normalising before computing the paired contrast. The important claim is that the sign, magnitude, and ranking of the reported contrast remain scorer-sensitive on fixed generations even after standardisation. We then run a compact second-generator probe (Qwen2.5, $n = 100$ fixed contexts per condition, one model only; not a population-level replication) and check whether the raw-contrast ordering is preserved.

Table 3: Fixed-generation metric fragility. Raw values are no-refinement–aggressive-refinement score differences on each scorer’s native scale and *are not directly comparable across scorers*. Mistral z/rank columns standardize each scorer over the same fixed rows before computing the paired contrast; bootstrap CIs are in the supplement. The first two rows are legacy zero-shot label proxies that do not consume the retrieved context (§3); the third is a context-conditioned local judge.

Scorer	Mistral raw	Mistral z	Mistral rank	Qwen2.5 raw
Legacy DeBERTa proxy (no context)	0.011	0.071 SD	0.022	0.015
Legacy roberta-large-mnli proxy (no context)	0.032	0.231 SD	0.062	0.046
Context-conditioned RAGAS-style judge	0.140	0.337 SD	0.085	0.134

Cross-scorer Pearson correlations on the same fixed Mistral rows are weak between the legacy DeBERTa proxy and the other two scorers (DeBERTa vs. **roberta-large-mnli**: $r = 0.259$; DeBERTa vs. RAGAS-style: $r = 0.182$) and moderate between the **roberta-large-mnli** proxy and the RAGAS-style judge ($r = 0.674$); supplement §6 shows the same ordering on the Qwen2.5 second-generator probe. The two NLI-style rows are legacy zero-shot entailment-label proxies (answer sentence as the sequence to be classified, retrieved context not consumed); the RAGAS-style row is context-conditioned by construction (the question, retrieved context, and answer are all passed to a local Mistral-7B that returns a support score with a short rationale). Scorer choice and scorer input format are therefore an audit axis, not hidden implementation trivia.

Scorer input format is itself an audit axis. To make the input-format claim concrete, we rescore a balanced subset of the same fixed-generation rows ($n = 200$ paired queries per condition, $n = 600$ total) under a context-conditioned NLI calling convention (premise = retrieved context, hypothesis = generated answer sentence) for both DeBERTa-v3 and **roberta-large-mnli**. Figure 1 reports the resulting baseline–aggressive-refinement contrasts. Under context-conditioned NLI the same two backbone models produce *sign-flipped* contrasts on the same fixed generations: the legacy and context-conditioned bootstrap intervals are disjoint

for both NLI backbones. Passing the retrieved context as premise to the same backbone weights changes the *sign* of the reported contrast at this audit cell.

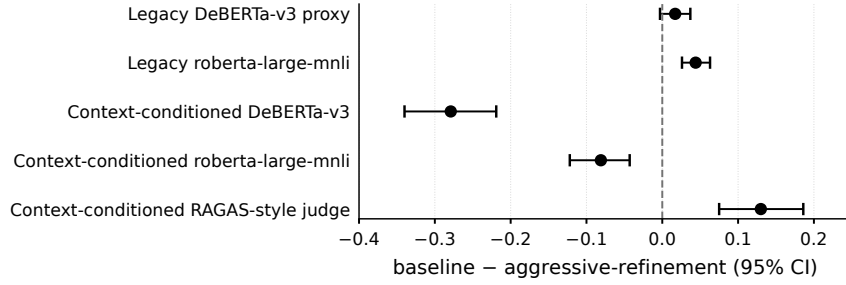


Figure 1: Same fixed generations, same NLI backbones, sign-flipped contrasts under different calling conventions (baseline—aggressive-refinement on the $n = 600$ balanced subset; 10,000-resample paired bootstrap 95% CIs).

Interpretation. Among the seven axes audited here, scorer choice is the largest movable quantity in the headline number, and it remains movable after standardisation; under proper context-conditioned NLI the two backbone NLI models produce sign-flipped contrasts on the same fixed generations. The $n = 99$ adjudicated calibration (§8.1) shows overlapping bootstrap CIs across scorers, so it does not settle a ranking either. Together these findings make scorer choice an audit axis, not an implementation detail: ControlledRAG requires at least two scorers, their input format (whether retrieved context is consumed), and a cross-scorer correlation to be reported. Scorer triangulation is not proof of correctness; it discloses how much of the reported number is scorer-dependent.

8 Human Calibration: Two Slices, Different Scorer Rankings

The human-calibration axis has two completed parts: the **typical-row calibration** ($n = 99$ stratified examples, two passes plus adjudication) and the **targeted disagreement slice** ($n = 100$ examples ranked by scorer disagreement, two raters plus adjudication). Both are load-bearing under ControlledRAG: the first estimates scorer alignment on the typical row distribution, the second stress-tests scorer alignment where automated scorers most disagree. The two slices reach *different* scorer rankings; that slice-dependence is the load-bearing finding.

8.1 Typical-row calibration ($n = 99$).

We sampled 99 rows from the 7,500 Mistral fixed-generation audit, stratified across no-refinement / aggressive-refinement / gated-refinement, and applied a two-pass annotation protocol with three labels (**supported**, **partially_supported**, **unsupported**) plus an adjudication step. Annotators used only the retrieved context. The two passes reach raw exact-match agreement 0.919 (91/99), Cohen’s $\kappa = 0.774$, with adjudicated distribution 79 / 16 / 4 (Table 4). Spearman correlations rank RAGAS-style above the **roberta-large-mnli** proxy above the legacy DeBERTa proxy, but the three bootstrap 95% CIs overlap, so the per-scorer ordering on this typical slice is suggestive only. The supplement (§12) contains Pearson and Kendall variants and the per-item verification trace.

8.2 Targeted disagreement slice ($n = 100$).

We construct a separate, targeted 100-example scorer-disagreement evaluation designed to stress-test metric-fragility claims where automated scorers disagree most sharply. The batch is *not* a random sample of the 7,500 rows: items are ranked by the absolute disagreement between the legacy DeBERTa proxy and the RAGAS-style judge, balanced across DeBERTa-high/RAGAS-low, RAGAS-high/DeBERTa-low, and near-threshold disagreement; the three retrieval conditions (no-refinement / aggressive-refinement / gated-refinement) are also balanced; items overlapping the $n = 99$ calibration are excluded. Two independent raters labelled all 100

items as **faithful** / **hallucinated** / **unclear** using only the retrieved context, and adjudication resolved the 12 rater-rater disagreements with documented notes.

Table 4: Completed human-evaluation slices: $n = 99$ typical-row calibration vs. $n = 100$ targeted disagreement slice. The two slices answer different audit questions and reach different scorer rankings; that slice-dependence is the load-bearing finding. Bootstrap 95% CIs are paired bootstrap from the verification scripts; full per-quantity CIs in supplement §12–§13.

Quantity / scorer	$n = 99$ typical-row calibration		$n = 100$ targeted disagreement	
	Value	95% CI	Value	95% CI
n adjudicated	99	—	100	—
Adjudicated label dist.	79/16/4	supp/part/unsup	74/26/0	faith/hall/unclear
Raw agreement	0.919	91/99	0.88	[0.81, 0.94]
Cohen’s κ	0.774	—	0.721	[0.588, 0.844]
Spearman, DeBERTa proxy	+0.103	[−0.080, +0.283]	−0.156	[−0.331, +0.032]
Spearman, roberta-large-mnli	+0.394	[+0.222, +0.543]	+0.446	[+0.293, +0.581]
Spearman, RAGAS-style	+0.549	[+0.380, +0.707]	+0.384	[+0.241, +0.523]
AUROC, DeBERTa proxy	—	—	0.398	[0.273, 0.514]
AUROC, roberta-large-mnli	—	—	0.794	[0.699, 0.879]
AUROC, RAGAS-style	—	—	0.718	[0.637, 0.793]

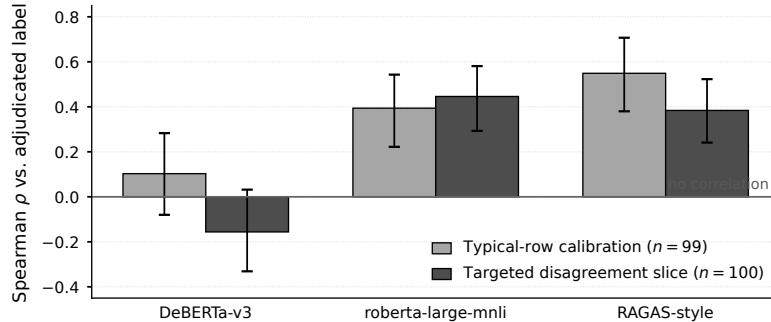


Figure 2: Scorer ranking against adjudicated humans is slice-dependent: RAGAS-style leads on typical rows; **roberta-large-mnli** leads on the disagreement slice, where the legacy DeBERTa proxy turns weakly negative. Error bars are bootstrap 95% CIs.

Interpretation. Figure 2 makes the inversion visually obvious: on the targeted disagreement slice **roberta-large-mnli** is the strongest-aligned scorer (Spearman $\rho = 0.446$, AUROC 0.794, AUPRC 0.929); RAGAS-style is second ($\rho = 0.384$, AUROC 0.718); the legacy DeBERTa proxy turns weakly negative ($\rho = -0.156$, AUROC 0.398). This is *not* the typical-row ordering. We do not claim any single scorer is universally best, and we do not treat the $n = 100$ slice as a population-level estimate. We treat the typical-row calibration plus an adversarial disagreement probe as a defensible minimum disclosure for this axis; finer protocols are an open question.

9 Threshold Transfer

Recovery is the dimensionless normalised contrast $R_\tau(d) = (\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{\text{v1}}(d)) / (\text{faith}_{\text{base}}(d) - \text{faith}_{\text{v1}}(d))$; **values can exceed 1 or be negative when the no-refinement–aggressive-refinement denominator is small or sign-inverted**, so the table is a transport diagnostic, not a universal utility score. A faithfulness claim that depends on a tuned operating threshold τ should report whether τ transfers across datasets. We sweep $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ for a CCS-gate retrieval intervention. For each tune dataset, τ^* is the threshold that maximises recovery on that dataset alone; diagonal recovery evaluates τ^* on the same dataset, while off-diagonal recovery evaluates it on the four other datasets.

Table 5: Threshold transfer summary. For each tune dataset, τ^* is the value of τ that maximises recovery on that dataset alone; “diagonal recovery” is recovery at τ^* on the same dataset; “off-diagonal mean” is the mean recovery at τ^* over the four *other* eval datasets. A large positive gap means the tuned τ^* does not transport. The full per-cell transfer matrix is in supplement §11.

τ tuned on	τ^*	Diagonal recovery	Off-diagonal mean	Gap (diag – off-diag)
PubMedQA	0.4	+1.452	+0.451	+1.002
NaturalQs	0.5	+1.401	+0.211	+1.191
SQuAD	0.3	+0.823	−0.141	+0.963
TriviaQA	0.4	+0.464	+0.698	−0.234
HotpotQA	0.3	−0.112	+0.093	−0.205

Tune-SQuAD and tune-HotpotQA both pick $\tau^* = 0.3$, and tune-PubMedQA and tune-TriviaQA both pick $\tau^* = 0.4$, so the off-diagonal columns coincide for those tune-dataset pairs in the full 5×5 matrix (supplement §9); this is a faithful consequence of the experiment, not a duplicate-row bug.

Interpretation. PubMedQA, NaturalQuestions, and SQuAD show a positive diagonal-vs-off-diagonal gap large enough to flag: tuning τ on one of them and transporting it loses most of the on-diagonal normalised contrast. The transfer table contains sign-changing cells, which is the load-bearing finding for the threshold axis.

10 Cost-Aware Baselines

Faithfulness numbers reported without latency and indexing cost are under-identified along the cost axis. This section is not a leaderboard; it is an identifiability test for deployment claims. We re-run a head-to-head on $n = 200$ SQuAD and $n = 200$ HotpotQA queries with three baselines: CRAG [Yan et al., 2024], gated-refinement (HCPC-v2), and a 2-level RAPTOR [Sarthi et al., 2024] index.

Table 6: Cost-aware head-to-head, $n = 200$ per (dataset, condition). Latency is end-to-end per query; indexing time is the offline cost of building the dataset’s retrieval structure. Faithfulness CIs are bootstrap 95% intervals; hallucination CIs are Wilson 95% intervals. RAPTOR-2L’s indexing cost is $17\text{--}22 \times$ the dense-baseline ($n = 200$ per cell).

Dataset	Condition	Faithfulness	Hallucination	Mean latency	Indexing
SQuAD	CRAG	0.698 [0.675, 0.722]	12.0% [8.2, 17.2]	1,440 ms	8.0 s
SQuAD	gated-refinement (HCPC-v2)	0.708 [0.685, 0.732]	12.5% [8.6, 17.8]	1,720 ms	8.0 s
SQuAD	RAPTOR-2L	0.710 [0.685, 0.735]	12.5% [8.6, 17.8]	1,714 ms	175.4 s
HotpotQA	CRAG	0.643 [0.627, 0.658]	10.5% [7.0, 15.5]	1,941 ms	8.0 s
HotpotQA	gated-refinement (HCPC-v2)	0.633 [0.617, 0.650]	12.5% [8.6, 17.8]	2,289 ms	8.0 s
HotpotQA	RAPTOR-2L	0.631 [0.615, 0.647]	13.0% [9.0, 18.4]	2,414 ms	134.0 s

Harness and cost accounting. This is a closed, zero-dollar local/free-tier harness: local Ollama/Mistral generation, Chroma indexing, MiniLM cross-encoders, no paid APIs. The CRAG reimplementation uses the documented evaluator/strip-refinement structure with live web search disabled, so its external-call count is zero. A live-web CRAG would need to report web-call count, API cost, and network latency separately. Self-RAG [Asai et al., 2024] is reported only in the supplement because its reflective decoding violates the fixed-context assumption.

At $n = 200$ per cell, faithfulness differences are inside CI overlap and the only axis with non-overlapping ranges is indexing cost ($17\text{--}22 \times$ for RAPTOR-2L). The cost axis is therefore the only axis that produces a defensible ranking on this cell, and ControlledRAG asks for it to be reported alongside the faithfulness table.

11 Discussion

What survives, by axis. Strength tags reflect evidence weight on this audit, not effect size.

- *Generator* [**WEAK**]: second-generator probe (Qwen2.5, $n = 100$, one model) preserves the metric-fragility ordering; not a panel claim.
- *Retriever* [**MEDIUM**]: matched HIGH>LOW CCS preserved across MiniLM/BGE/E5 (supp. §4); fixed generations unchanged.
- *Context structure* [**STRONG**]: matched CCS intervention null (-0.002 , paired Wilcoxon $p = 0.628$); span and CCS are complementary diagnostic axes.
- *Scorer* [**STRONG**]: raw contrasts 0.011/0.032/0.140 on the same fixed generations; context-conditioned NLI sign-flips both backbones with disjoint bootstrap CIs (§7).
- *Human calibration* [**STRONG**]: $n = 99$ ($\kappa = 0.774$) ranks RAGAS-style first; $n = 100$ ($\kappa = 0.721$) ranks `roberta-large-mnli` first and the legacy DeBERTa proxy weakly negative. Slice-dependent.
- *Threshold* [**MEDIUM**]: 5×5 transfer table has sign-changing cells; no bootstrap CIs.
- *Cost* [**WEAK**]: faithfulness CIs overlap at $n = 200$; the indexing-cost gap (17–22× for RAPTOR-2L) is the clear signal.

Coherence-preserving and random-noise slopes differ in point estimate, but no significance test was run; see supplement §10. Scope limits on the empirical demonstration are categorised in §12.

The seven axes are load-bearing in different proportions. ControlledRAG asks for all seven to be reported because under-identification on any single one can move the headline; the weight of evidence on a given axis (the [**STRONG**] / [**MEDIUM**] / [**WEAK**] tags above) reflects this audit, not a property of the standard. In this audit, all seven axes are load-bearing, but in different proportions.

12 Limitations and Ethics

Audited surface. Audited cells are mostly short-answer extractive QA on SQuAD with 7B-class generators; the matched-context audit is SQuAD-only; the long-form QASPER/MS-MARCO probe has 40 distinct questions and is supplement-only. Self-RAG [Asai et al., 2024] runs in a harness mismatched with the audit’s fixed-context protocol and is supplement-only. Stronger embeddings preserve the matched HIGH>LOW CCS ordering, but the fixed-generation null is unchanged.

Human evaluation. The typical-row calibration ($n = 99$) has overlapping bootstrap CIs across scorers; the targeted disagreement slice ($n = 100$) is adversarial-by-construction. Both arms rest on two raters plus adjudication, not a broad annotator pool, and neither slice settles a global scorer order. The slice-dependence of the ranking is the load-bearing finding.

Scope of replication. The second-generator probe is one local Qwen2.5 model at $n = 100$ fixed contexts per condition (a sanity check, not a panel replication); the context-conditioned NLI rescoring is on a $n = 600$ balanced subset, not the full 7,500 (sufficient to demonstrate input-format sign-flipping, not a full-table replacement).

Ethics. Public datasets and free-tier or local compute only; the annotation arms cover short-answer support judgments on public content (no PII, no sensitive content); annotator identities are not stored. Review artifact anonymized; public links follow acceptance.

These limits narrow the breadth of the empirical demonstration but do not threaten the methodological claim, because the seven-axis decomposition is motivated by the disclosure principle in §4 and not by any single empirical cell.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.

- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–163, 2024.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2024.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. TRUE: Re-evaluating factual consistency evaluation. *NAACL*, 2022.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2567–2577, 2019.
- Vladimir Karpukhin, Barlas Öguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergei Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781, 2020.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. In *Transactions of the Association for Computational Linguistics (TACL)*, 2022.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- Jiwei Li and Eduard Hovy. A model of coherence based on distributed sentence representation. *Proceedings of EMNLP*, 2014.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 2024.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.585. URL <https://aclanthology.org/2024.acl-long.585/>.
- Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.
- Chanhee Park, Hyeonseok Moon, Chanjun Park, and Heuiseok Lim. MIRAGE: A metric-intensive benchmark for retrieval-augmented generation evaluation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2883–2900, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-naacl.157. URL <https://aclanthology.org/2025.findings-naacl.157/>.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, 2016.
- Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Jiayang Cheng, Cunxiang Wang, Shichao Sun, Huanyu Li, Zizhao Zhang, Binjie Wang, Jiarong Jiang, Tong He, Zhiguo Wang, Pengfei Liu, Yue Zhang, and Zheng Zhang. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation. *arXiv preprint arXiv:2408.08067*, 2024. doi: 10.48550/arXiv.2408.08067. URL <https://arxiv.org/abs/2408.08067>.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.20. URL <https://aclanthology.org/2024.naacl-long.20/>.
- Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2401.18059>.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2023.
- Manveer Singh Tamber, Forrest Sheng Bao, Chenyu Xu, Ge Luo, Suleman Kazi, Minseok Bae, Miaoran Li, Ofer Mendelevitch, Renyi Qu, and Jimmy Lin. Benchmarking LLM faithfulness in RAG with evolving leaderboards. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 799–811, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-industry.54. URL <https://aclanthology.org/2025.emnlp-industry.54/>.
- Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2024.
- Fangyuan Xu, Weijia Shi, and Eunsol Choi. RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.04408>.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 2564–2579, 2024.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, 2018.

NeurIPS Paper Checklist

1. Claims

Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The paper title and contributions frame the work as an evaluation-methodology standard (*ControlledRAG: A Seven-Axis Reporting Standard for RAG Faithfulness Claims*). The abstract and introduction explicitly disclose: the matched-context CCS null; the small and seed-sensitive scaled headline (no-refinement / aggressive-refinement / gated-refinement, prior names HCPC-v1 / HCPC-v2); the fixed-generation scorer fragility and context-conditioned NLI sign-flip; the slice-dependent scorer–human ranking between the typical-row calibration ($n=99$) and the targeted disagreement slice ($n=100$); the threshold-transfer instabilities; and the cost-axis dominance of the head-to-head. The contribution is the standard, not a new method or leaderboard.

2. Limitations

Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §12 is grouped into three categories: *audited surface* (mostly short-answer SQuAD, SQuAD-only matched-context, supplement-only 40-question long-form probe, mismatched Self-RAG harness); *human evaluation* (typical-row calibration $n=99$ with overlapping bootstrap CIs, targeted disagreement slice $n=100$ adversarial-by-construction, two raters plus adjudication on each); *scope of replication* (one local Qwen2.5 second-generator probe at $n=100$, context-conditioned NLI rescoring on the $n=600$ balanced subset not the full 7,500). The section ends with the explicit statement that these limits do not threaten the methodological claim because the seven-axis decomposition is motivated by the disclosure principle in §4.

3. Theory Assumptions and Proofs

For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper makes no formal theoretical claims requiring proofs. All claims are empirical or definitional.

4. Experimental Result Reproducibility

Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §3 specifies the pipeline, embedder, indexer, reranker, generator, decoding settings, sample sizes, seed list, and statistical protocol. The supplement (§1 Pre-Specification, §2 per-seed scaled detail, §6 scorer implementation) provides full reproduction details. The artifact ships per-query CSVs and scripts. *Reproducibility scope:* `run_all_analysis.sh` is end-to-end on a laptop in under two minutes for the verification path (it regenerates every analysis table in the main paper from the frozen per-query CSVs in `results/revision/` and `human_eval_final/`). Re-running the original *generation* and *scoring* pipelines (matched-context generation, scaled audit, second-generator probe, context-conditioned NLI rescoring) is a separate path that needs local Ollama plus a CPU/MPS or free-tier GPU notebook and takes hours; supplement §15 documents that path. The two-minute verification path covers all numerical claims in the main paper.

5. Open access to data and code

Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: The reviewer-facing anonymized artifact (supplementary ZIP) contains all per-query CSVs, the human-evaluation final files, analysis scripts, the pre-specification log, the source trace, and a reproducibility README with verification commands. Public release links and DOIs will be added at acceptance.

6. Experimental Setting/Details

Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.)?

Answer: [Yes]

Justification: The audit holds the entire RAG pipeline fixed and varies only the audit axis under test. Embedder, indexer, reranker, refinement policy, generator, and decoding settings are listed in §3; sample sizes and seeds are listed for every audited cell; the matched-context construction (similarity gap ≤ 0.02 , CCS gap ≥ 0.20) is given.

7. Experiment Statistical Significance

Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every reported contrast is paired with a 10,000-resample bootstrap CI, paired Wilcoxon p -value where applicable, Cohen’s d_z for paired contrasts, Wilson 95% CIs for binary rates, and bootstrap CIs for the cross-scorer correlations. Multiple-comparison handling: only the matched-context test is treated as the pre-specified primary; other paired contrasts are descriptive across audit cells, as disclosed in §3.

8. Experiments Compute Resources

For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The supplement §Resume and Compute Notes specifies the local Ollama / Apple-silicon MPS / free-tier Kaggle T4 hardware, the per-fix runtime envelopes, and the analysis-only verification path (`run_all_analysis.sh`, <2 min on any laptop). No paid APIs are required.

9. Code Of Ethics

Does the research conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work uses only public datasets, free or local compute, no human-subjects data beyond two short-answer support-judgment annotation passes by adult raters on public-dataset content. No personally identifiable information, sensitive attributes, or facial / biometric content is involved. No deceptive practices. Annotator identities are not stored in the artifact.

10. Broader Impacts

Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: ControlledRAG’s intended impact is to make RAG faithfulness claims more honestly auditable, which reduces the risk that deployed RAG systems are fielded based on under-identified single-cell numbers (negative impact: false confidence in deployed medical / legal RAG; the standard is meant to mitigate that). The work is descriptive; it does not introduce a new generative model that could itself be misused.

11. Safeguards

Does the paper describe safeguards that have been put in place for responsible release of data or models

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12. Licenses for existing assets

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Justification: The paper cites SQuAD, PubMedQA, HotpotQA, NaturalQuestions, TriviaQA, QASPER, MS-MARCO, and the open embedder/cross-encoder/NLI models used. Each is used under its published research-use license. No proprietary asset is used.

13. New Assets

Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The new released assets are (i) the ControlledRAG seven-axis reporting standard (§4) plus a filled-in reporting-checklist template (`controlledrag_reporting_checklist.md`, distinct from this mandatory NeurIPS checklist), (ii) per-query CSVs for every audited cell, (iii) the typical-row calibration set ($n=99$), (iv) the targeted disagreement slice ($n=100$, two raters plus adjudication, 74/26/0 adjudicated distribution), and (v) the analysis scripts. Documentation: `README_REPRODUCE.md`, `SOURCE_TRACE.md`, `artifact_manifest.md`, `CLAIMS_AUDIT.md`, and the supplement Source-Trace section.

14. Crowdsourcing and Research with Human Subjects

For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [Yes]

Justification: The two adjudicated human-annotation arms ($n=99$ typical-row calibration; $n=100$ targeted scorer-disagreement evaluation) are documented with their exact label scheme, rater instructions, decision codebook, and adjudication template; these documents are shipped in the artifact under `human_eval_final/`. No crowdsourcing platform was used; no compensation was paid; the raters were not subject to risk.

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Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The annotation tasks are short support/faithfulness judgments on public-dataset content (no personally identifiable information, no sensitive attributes, no risk-of-harm content). Per common interpretations of the relevant human-subjects regulations, this evaluation does not constitute research with human subjects requiring IRB review. The annotators were informed of the purpose of the labelling.

16. Declaration of LLM usage

Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: A local Mistral-7B (via Ollama) is used as both the audited generator and as the

RAGAS-style local judge in the metric-fragility audit. A local Qwen2.5-7B is used as the second-generator replication. Their use, decoding settings, and the fact that the RAGAS-style judge is itself an audit input (not a ground-truth oracle) are disclosed in §3 and §7. No LLM is used to generate the paper text or the analysis tables; LLMs are research artifacts, not co-authors.